

# The famous case of Netscape

What separates successful software systems from not-so-successful systems?

# The rise and fall of Netscape

# Case of Netscape Communications

- One of the most talked about companies for its innovation.
- Flagship product – Netscape Navigator (1990)
  - A proprietary web browser with a huge user base and market dominance.
  - Had very good features, commercial licensing, promotion from ISPs
  - Was well positioned to take advantage of Internet revolution in late 1990s.
- April – Dec 1994, released Navigator 1.0
  - For three platforms, instant success on Windows.
  - Innovations included – frames, cookies, SSL (secure sockets layer) protocol for secure communications.
  - Introduced Javascript in its version 2.0

# Rise of Netscape

- Microsoft realized the importance of having its own web browser.
  - Began development of IE in Oct 1994, and launched it in Aug 1995
  - Competition was one-sided, No effect on Netscape.
- During the same time,
  - Netscape was experimenting with web-based mini OS (Constellation) to enable users to access, edit and share files independent of hardware and OS they used.
    - This was a big threat to Microsoft.
- Netscape released its version 4.0
  - E-mail client, messenger, collaborative threaded discussion, audio and video conferencing, group calendaring and scheduling.
  - This package was called Communicator
    - 3 millions LOC, around 120 developers
    - This was the period of rapid growth of Netscape.

# Fall of Netscape (The Browser War)

- Microsoft could not catch up Netscape till IE 3.0
  - Started full fledged campaign to increase its share
  - With IE 4.0, it appeared like a serious competition
  - With IE 5.0, Microsoft surpassed Netscape
  - Finally, the war was won by Microsoft.
- What is interesting for us to look in this?
  - Netscape bundled a lot of functionality with its 4.0
  - Became spaghetti code because all developers were just focused on adding a lot of functionality on war footing.
    - What was earlier **fast and robust**, **became slow, buggy and crash prone**.
  - Netscape abandoned its effort to release 5.0
    - Threw open the code base of 5.0, for which there were no takers.
  - Netscape started all over again with Communicator 6.0 with a new code base
    - Later released version 7.0 before its disappearance.

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The usage share of Netscape had fallen from over 90 percent in the mid-1990s to less than one percent by the end of 2006

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Shortly before its acquisition by AOL, Netscape released the source code for its browser and created the Mozilla Organization to coordinate future development of its product

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Industry experts feel that  
Microsoft is not solely  
Responsible for the fall of  
Netscape

**Netscape scripted its own end**

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**What does the story of Netscape convey?**

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The result of attempting to scale up without  
Planning for such growth.

Adhoc software development cannot win in the  
Long run.

**Lesson: Software systems developed without  
A solid architecture and design cannot scale  
Up to meet the growing demands.**